

SINETHEMBA MVELASE

Artificial Intelligence Specialist | AI Engineer | Generative AI & LLMOps

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EXECUTIVE SUMMARY

Results-driven **Artificial Intelligence Specialist** with hands-on expertise in **Generative AI, Large Language Models (LLMs), LLMOps, Natural Language Processing (NLP), and Conversational AI**, complemented by a strong foundation in Mathematics, Physics, and Enterprise Technology Management. Proven ability to architect and deploy production-grade AI systems on **AWS Bedrock, Anthropic Claude, Google Vertex AI, and BigQuery**, translating complex business requirements into scalable, cloud-native AI solutions.

Currently serving as a **Discover Graduate Trainee (Technology Management & AI)** at **Vodacom Financial Services**, supporting C-suite executives across Service Management, PI Delivery, Risk, Technology Assurance, and AWS Cost Optimization, while independently designing multilingual AI products (**KasiKash, CAPS AI Tutor**) that leverage **RAG pipelines, prompt engineering, voice AI, and WhatsApp/USSD conversational interfaces** to serve underserved African markets.

Combines executive-level business acumen with deep technical fluency in Python, TypeScript, cloud engineering, and CI/CD — uniquely positioned to bridge AI engineering, advanced analytics, and enterprise strategy.

CORE COMPETENCIES

- AI & Machine Learning:** Generative AI • LLMs • Prompt Engineering • Retrieval-Augmented Generation (RAG) • Conversational AI • Chatbot Development • NLP • LLMOps • AI Automation • Machine Learning • Voice AI • Multilingual NLP
- Cloud & AI Platforms:** AWS • Amazon Bedrock • Anthropic Claude • Google Cloud Platform (GCP) • Vertex AI • BigQuery • Supabase • Vercel • Render
- Programming & Engineering:** Python • TypeScript • JavaScript • SQL • Node.js • Express.js • React 19 • PostgreSQL • Prisma • Redis • Socket.IO • REST APIs
- Data & Analytics:** Advanced Analytics • Data Science • Business Intelligence • Power BI • Excel (Advanced) • Data Visualization • Exploratory Data Analysis • Executive Reporting
- DevOps & LLMOps:** Git • GitHub • GitHub Actions • CI/CD Pipelines • Model Deployment • Cloud Cost Optimization • Agile Delivery
- Business & Governance:** Technology Assurance • Risk Management • Project Governance • Executive Presentations • Stakeholder Engagement • ITIL 4

TECHNICAL SKILLS MATRIX

Domain	Technologies & Tools	Proficiency
Generative AI & LLMs	AWS Bedrock, Anthropic Claude, Vertex AI, Prompt Engineering, RAG	Advanced
Conversational AI	Chatbots, Voice AI, Web Speech API, WhatsApp API, USSD Integration	Advanced
NLP	Multilingual Processing (English, isiZulu, Sesotho), Text Analytics	Advanced
Cloud Platforms	AWS, GCP, BigQuery, Vertex AI, Supabase	Advanced
Programming	Python, TypeScript, JavaScript, SQL	Advanced
Data Analytics	Power BI, Excel, BigQuery, Data Visualization	Advanced
DevOps / LLMOps	GitHub Actions, CI/CD, Vercel, Render	Intermediate–Advanced
Backend Engineering	Node.js, Express, PostgreSQL, Prisma, Redis, Socket.IO	Intermediate
Frontend Engineering	React 19, Vite, Tailwind CSS, Framer Motion	Intermediate

PROFESSIONAL EXPERIENCE

Discover Graduate Trainee — Technology Management & AI Feb 2025 – Present
Vodacom Financial Services — Johannesburg, South Africa

Selected for Vodacom's flagship Discover graduate programme, embedded within senior technology leadership to drive AI-led decision-making, executive reporting, and cloud cost governance across the Financial Services division.

- Partner with **C-suite executives and senior management** across Service Management, PI Delivery, Risk Management, Technology Assurance, and Steering / Management Committees to deliver AI-informed business insights.
- Lead **AWS Cost Optimization** initiatives, analysing consumption patterns and recommending governance actions that improve cloud spend efficiency across the FinServ technology estate.
- Design and deliver **executive-level dashboards, PowerPoint decks, and analytical reports** consumed by ExCo and Steering Committees for strategic decision-making.
- Apply **AI and advanced analytics** to operational data, automating reporting workflows and increasing insight turnaround for cross-functional technology initiatives.
- Contribute to **project governance, risk registers, and technology assurance frameworks**, aligning delivery with **ITIL 4** best practice.
- Support strategic AI adoption initiatives, evaluating **Generative AI** and **LLM** use cases for internal automation and customer experience.

Reporting Analyst Intern Jun 2024 – Jan 2025
Vodacom Financial Services — Johannesburg, South Africa

- Developed and maintained **Power BI dashboards** and **advanced Excel models** used by business leaders for performance monitoring.
- Delivered recurring and ad-hoc **business performance reports**, translating raw operational data into commercial insight.
- Automated repetitive reporting tasks, reducing manual effort and improving accuracy of executive analytics.
- Collaborated with finance, technology, and operations teams to standardise KPIs and reporting cadences.

Data Scientist (Project-Based) Cross-Border Engagement
Vodafone Portugal — Data Science Project

- Engineered SQL queries on **Google BigQuery** to extract, clean, and transform large-scale enterprise datasets.

- Performed **exploratory data analysis (EDA)** and feature preparation on **Google Cloud Platform**, leveraging **Vertex AI** for ML workflows.
- Produced analytical reports and business insights that informed data-driven decisions for the Vodafone Portugal team.
- Strengthened core competencies in cloud-native data science, GCP tooling, and enterprise analytics.

AI PROJECTS PORTFOLIO

KasiKash — AI-Powered Multilingual Financial OS for Township SMEs
Founder & Lead AI Engineer

- An AI-driven, voice-first financial operating system enabling township small businesses to run bookkeeping, credit, and stokvel management in their own languages.
- Architected a **multilingual Conversational AI platform** supporting **English, isiZulu, and Sesotho**, using the **Web Speech API** and NLP pipelines to convert spoken transactions into structured bookkeeping records.
 - Built "**Skorokoro**", an AI-powered customer debt tracking module, and "**KasiScore**", a proprietary **alternative credit scoring engine** for the unbanked and underbanked.
 - Implemented **stokvel contribution management** with automated reconciliation and financial intelligence.
 - Delivered end-to-end using **React 19, TypeScript, Vite, Tailwind CSS, Framer Motion, Supabase (PostgreSQL)**, with **GitHub Actions CI/CD** deployed on **Vercel**.
 - **AI Stack:** Voice AI, NLP, multilingual language processing, financial intelligence automation.

CAPS AI Tutor — Conversational LLM Tutor for South African Learners
Founder & AI Engineer

- A Generative AI education assistant aligned to the South African **CAPS curriculum**, delivering personalised tutoring at scale via chat, WhatsApp, and USSD.
- Engineered an **LLM-powered tutoring agent** on **AWS Bedrock** using **Anthropic Claude**, tuned via **prompt engineering** and grounded on curriculum content.
 - Integrated **WhatsApp and USSD channels**, enabling low-bandwidth, feature-phone access for learners across South Africa — a true production **Conversational AI / Chatbot** deployment.
 - Implemented **multilingual NLP support** for South African languages, personalised learner Q&A, and adaptive tutoring flows.
 - Deployed with **Python, GitHub Actions (CI/CD), Vercel, and Render**, applying **LLMOps** practices for prompt versioning and cost-controlled inference.
 - **AI Stack:** Generative AI, LLMs, Claude on Bedrock, RAG-ready architecture, Chatbots, NLP.

Enterprise Delivery Management Platform — Takealot Drivers
Full-Stack Engineer

- Enterprise-grade delivery operations platform for last-mile drivers, dispatchers, and administrators.
- Built driver **trip tracking, cash-on-delivery verification, real-time notifications, and dispatch management** modules.
 - **Frontend:** React, TypeScript, Vite, Tailwind, React Query, Zustand.
 - **Backend:** Node.js, Express.js, PostgreSQL, Prisma, Redis, Bull Queue, JWT Authentication, Socket.IO for real-time events.
 - Demonstrated capability in **cloud architecture, real-time systems, and production backend engineering** — foundational to deploying scalable AI services.

EDUCATION

- Enterprise and Technology Management**
NWU Business School • 2024
- BSc Mathematics and Physics**
University of Johannesburg • 2019 – 2023

KEY STRENGTHS

- Deploying **Generative AI and LLM** applications on **AWS Bedrock, Claude, and Vertex AI**.
- Designing **Conversational AI and Chatbots** for multilingual, low-bandwidth African contexts.
- Bridging **AI engineering** with **executive reporting, governance, and cloud cost optimization**.
- Rapid delivery via **CI/CD, GitHub Actions, Vercel, and Render** with modern **LLMOps** hygiene.
- Strong mathematical foundation (BSc Mathematics & Physics) underpinning ML work.

CERTIFICATIONS

- Machine Learning Specialization**
DeepLearning.AI / Stanford University (Andrew Ng)
- ITIL 4 Foundation**
IT Service Management

LANGUAGES

- English — Professional
- isiZulu — Native
- Sesotho — Conversational

REFERENCES

Available on request.